Experiment #1 -
Testing for Sequential Linear Separability

https://colab.research.google.com/drive/1glI17EJJEdi8scGCjhpa42ZmpCsZ4CVU?usp=sharing

What is Done?:

- Tests whether real data (MNIST digits **0, 3, 8**) satisfy **Sequential Linear Separability (SLS)**
- Empirically checks if an ordering of classes exists such that each class can be separated one-vs-all **after previous collapses**

Methodology:

- Load MNIST dataset and **filter to three classes (0, 3, 8)**
- Flatten images into **784-dimensional vectors**
- Compute **class means** for each digit
- Test SLS using **iterative one-vs-all linear SVMs (LinearSVC)**:
    - At each step, treat one class as target and remaining as background
    - Fit a separating hyperplane
    - Remove the separated class and repeat on the reduced set
- Evaluate separability orderings via **SVM feasibility**, not neural training
- Visualize distributions and separation behavior for interpretability

Results:

- MNIST digits 0, 3, 8 are **NOT sequentially linearly separable** in raw pixel space.
- Most MNIST digit combinations are **NOT sequentially linearly separable**.

Experiment #2 - Visualizing the Cones - Part 1

(until PCA of Traing Data + Overlapping Cones)

https://colab.research.google.com/drive/1owO-VAnCmoNxCXqZ91DN30ZMfRnAbYNi?usp=sharing

What is Done?:

- Trains a **standard ReLU neural network** on MNIST digits (**0, 3, 8**) according to **CE24**.
- Extracts **learned weights and biases** to interpret the trained network via **cumulative parameters**.
- Begins **visualizing the truncation geometry (cones)** corresponding to trained layers.

Methodology:

- Filter MNIST to digits **0, 3, 8** and flatten images into **784-dimensional vectors**
- Train a **ReLU feedforward network** with architecture motivated by **CE24 Theorem 5.2**:
  - $L=Q+1$ layers
  - Constant hidden width $M \geq Q$
  - Optimize using **SGD with momentum** and loss
- After training:
  - Extract layerwise weights W_ℓ and biases b_ℓ
- Compute **cumulative parameters** $W^{(\ell)}$, $b^{(\ell)}$ via chained matrix products
- Use cumulative parameters to **prepare cone / truncation-map visualization**
- Hand-derived calculations included to verify correctness of cumulative expressions

Results:

- Network successfully **trains to low classification error**
- Learned parameters can be **reinterpreted geometrically** as truncation maps
- Confirms that **trained networks implicitly realize CE24-style constructions**
- Sets up Part II:
  - Explicit cone visualization
  - Comparison between **trained vs analytic truncation geometry**

Experiment #3a -
Visualizing the Cones - Part 2

(until PCA + Synthetic Data Generation + Truncation Map Visualization + Hyperplane Experiment)

https://colab.research.google.com/drive/1DySk1rDp_Lh1OFsQH6mcoxj0euNSfDzO?usp=sharing

What is Done?:

- Visualizes how truncation maps affect MNIST classes (**0, 3, 8**) using:
  - Global PCA
  - Local PCA
  - Barycentric (mean-difference) projections
  - Linear Discriminant Analysis (LDA)
- (Dataset A) Create a 3-class clustered Gaussian dataset in R^3 to serve as a SLS example.
- (Dataset B) Construct a non-clustered 3-class dataset that remains sequentially linearly separable.
- (Dataset C) Build a two-class dataset that violates both SLS and concentric structure.
- (Dataset D) Create two concentric classes (disk and ring) to model non-linearly separable data.
- I began a hyperplane experiment which can be found at the end of the attached Google Colab document, but did not any novel progress, so I moved on.

Methodology:

- Aggregate full training data into a single matrix X_{train}
- Compute **cumulative truncation maps** using CE24 definition:
  - $\tau_{W,b} = W^{+}(\mathrm{ReLU}(Wx+b)-b)$
- For each hidden layer:
  - Apply truncation map using **pseudoinverse of cumulative weights**
  - Store truncated representations layer-by-layer
- Visualization strategies:
  - **Global PCA**: project all truncated layers into the same PCA basis fitted on original data
  - **Local PCA**: fit PCA independently on each truncated dataset
  - **Barycentric projection**:
    - Compute class means in truncated space
    - Project data onto mean-difference directions
  - **LDA projection**:

        - Fit LDA on original data
        - Project truncated data into the same discriminant subspace
- Final layer logits visualized directly in R^Q
- (Dataset A) Generate well-separated Gaussian blobs and verify SLS via iterative one-vs-all linear SVMs.
- (Dataset B) Generate elongated class distributions aligned with distinct separating directions and test SLS via LinearSVC.
- (Dataset C) Place one class inside a polyhedral cone and the other outside, then test SLS failure explicitly.
- (Dataset D) Generate filled-disk and annular distributions and embed them in R^3.

Results:

- Shown as images in experiment, this is a visual task, so hard to describe what is going on.
- (Dataset A, B, C, D) The dataset was created successfully, and the result is known by project participants.

Experiment #3b - Visualizing the Cones - Part 2

(adding No Limit to Epochs + Plateau Condition)

https://colab.research.google.com/drive/1xltQWLZ_nE5PpxmCV_NUVJBsqASBH7DI?usp=sharing

What is Done?:

- Training is no longer constrained by a fixed epoch count and instead stops based on **loss convergence behavior**.

Methodology:

- Introduce **dual stopping criteria**: early stop when loss crosses a target threshold or when improvement falls below a plateau tolerance for a fixed window.

Results:

- Allowing unconstrained training with plateau-based stopping enables some datasets to **reach zero training loss and 100% accuracy**, which was not observed under fixed-epoch limits.

Experiment #4 -
Zero Loss Global Minima Experiments w/ Dataset C

https://colab.research.google.com/drive/1siWa1DOK8KJFaNUCnA84rYjmv_GNraWo?usp=sharing

What is Done?:

- Introduces an **explicit, hand-constructed ReLU network** (no training) that achieves **zero loss on Dataset C**, followed by a **stability analysis** of this zero-loss solution under controlled parameter perturbations.

Methodology:

- Define **polyhedral truncation cones** by choosing **apex points** and generating vectors aligned with the dataset geometry.
- Compute each cone's **cumulative truncation map** using the pseudoinverse relation $W=V^{+}$, $b=-Wp$ (Ewa25 Lemma 2.1)
- Stack multiple truncation maps to sequentially collapse each class to a distinct apex
- Convert the cumulative parameters to **layerwise ReLU weights** via CE24's cumulative-to-layerwise formulas (Eq. (5)–(6))
- Append a final affine layer mapping the collapsed apices to one-hot class labels, finally ending up with an **explicit ReLU network with zero training loss at initialization**.
- Verify zero loss at initialization
- Then, apply **controlled normalized perturbations** to all network parameters:
    - First along a single direction (λ-perturbations)
    - And then over a two-dimensional orthonormal basis (α,β); followed by SGD to test whether training **returns to the same zero-loss solution**.

Results:

- The explicit construction achieves **exact zero loss and perfect accuracy.**
- When you slightly change the network parameters away from the exact zero-loss construction, **training does not break**.
- Instead, **SGD naturally pulls the parameters back** to a solution with **perfect classification**.
- This happens not just for tiny changes, but across a **whole region of parameter space**, not a single fragile point.

Experiment #5 -
Zero Loss Global Minima Experiments w/ Multiple Datasets

https://colab.research.google.com/drive/1JpWZ-HBiT2TQyMp1vx5n3Q0GD1NOwRXe?usp=sharing

Important Note: In this case, *Multiple Datasets,* means the addition of Dataset D into the experiment, as Dataset A is still being worked on. Additionally, Dataset D was originally created using two embedded rings, but this isn't correct as the rings have to be initialized according to generating vectors according to CE24, so I created * **Addition to 3a. Generating Dataset Type D Using Vectors and Simplex.**

(https://colab.research.google.com/drive/1Xze1e2ce0UEICwsVISOFDjxBd4F5QgO5?usp=sharing)

Geometric setup:

- Construct a 2D simplex S (triangle), lift it to R^3, and choose a point p a fixed distance off the simplex plane; define cone generators $v_i = p - s_i$ from each simplex vertex.

Class separation by containment:

- Sample Class 0 strictly inside the simplex using barycentric coordinates (ensuring positive weights), and sample Class 1 strictly outside the simplex by drawing points in an region around the centroid.

Lifted separability mechanism:

- Embed all data into R^{n+1} so that points inside the simplex lie in the negative cone generated by $\{v_i\}$, while outside points lie outside it; allowing for a truncation-map-based zero-loss ReLU construction despite non-SLS geometry.

> Now back to creating the zero-loss construction for Dataset D, which is the purpose of this Google Colab Notebook… *I will just include* ***Methodology*** *to avoid redundancy.*

Methodology (according to Corollary 3.2 / Ewa25):

- **Sequential cone-based collapse via truncation maps:**
  - Two truncation maps are explicitly constructed using Lemma 2.1 by defining cone, generators V1=$[v_1,v_2,v_3]$ and V2=$[\hat{v}_1, \hat{v}_2, \hat{v}_3,]$ with distinct apices p and p_2; the corresponding cumulative parameters (W_{cum}, b_{cum}) collapse each class to a unique point despite non-SLS geometry.
- **Explicit affine separation of collapsed apices:**
  - After sequential collapse, a final linear layer (CE24 Theorem 6) maps the distinct apices $\{p,p_2\}$ to one-hot labels using a closed-form pseudoinverse solution, guaranteeing zero cross-entropy loss without training.
- **Exact layerwise realization:**
  - The cumulative parameters are converted into layerwise ReLU weights using CE24 equations (5)–(6), producing a ReLU network that provably achieves zero loss on non-SLS data.

Experiment #6 - Suboptimal Minima Stability Study `MOST RECENT CODE (1/14/26)`

https://colab.research.google.com/drive/1j1aQEkqcnUxqVmgdrm6f_AixRQSacBSc?usp=sharing

Important Note: Being a work in progress, this experiment currently only has the zero-loss construction for Dataset A, and is yet to begin the **Suboptimal Minima Stability Study.**

What is Done?:

- An explicit deep ReLU network is analytically constructed to achieve zero loss on Dataset A by sequentially collapsing each class to a distinct point and then separating those points linearly.

Methodology:

- For each class, a truncation cone is defined by selecting an apex and generator vectors; construct them (one truncation cone per class) using Lemma 2.1, with each apex placed *in front of* the class mean along the centroid direction; compute cumulative (W,b) via pseudoinverse.
- Diagnose cone orientation by comparing the cone axis (sum of generators) with the desired collapse direction, then **tilt generators** to correct misalignment while preserving cone structure.
- Apply sequential truncation maps to collapse each class to a distinct apex, followed by a linear map (CE24 Theorem 6) sending apices to one-hot labels; convert cumulative parameters to exact layerwise ReLU weights.

Results:

- ***(Work in progress)…*** we are currently working on fine-tuning the construction and achieving zero-loss & 100% accuracy with MSE as the loss function.
